SAPIENZA UNIVERSITY OF ROME

Course: **Quantitative Financial Modeling**

Stochastic Asset Dynamics and Derivatives Pricing

Derivation of the Black-Scholes PDE and the Merton Extension

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Abstract

This project provides a theoretical framework for stochastic asset dynamics and derivatives pricing, under the assumptions of frictionless markets and the strict absence of arbitrage. **Part A** begins by deriving *Itô's Lemma* in one dimension, highlighting the critical role of quadratic variation. Leveraging this stochastic chain rule, we construct a self-financing hedged portfolio to dynamically eliminate market risk, thereby extracting the foundational *Black–Scholes Partial Differential Equation* (PDE). **Part B** explores the Merton extension for assets paying a continuous dividend yield. It demonstrates how continuous cash outflows necessitate a localized structural adjustment to the drift parameter, ultimately deriving the modified pricing PDE under the equivalent martingale measure.

1. PART A: Derivation of Itô's Lemma in One Dimension

1.1 Setup and Notation

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a filtered probability space supporting a Standard Brownian motion $\{W_t\}_{t\geq 0}$. The stochastic process $X = \{X_t, t \geq 0\}$ that solves the **stochastic integral equation**

$$X_t = X_0 + \int_0^t \mu(s, X_s)\, ds + \int_0^t \sigma(s, X_s)\, dW_s \tag{1}$$

is called an **Itô process**. Its equivalent **stochastic differential equation (SDE)** form is:

$$dX_t = \mu(t, X_t)\, dt + \sigma(t, X_t)\, dW_t \tag{2}$$

where $\mu(t, X_t)\, dt$ is the deterministic *drift* (finite-variation term) and $\sigma(t, X_t)\, dW_t$ is the noisy martingale term — the only source of uncertainty, driven by weighted Brownian increments σ.

1.2 Derivation of Itô's Lemma

Itô's Lemma is the stochastic counterpart to the chain rule. Since Brownian motion is nowhere differentiable, possesses infinite total variation but finite quadratic variation, classical calculus fails and a second-order Taylor expansion becomes necessary. This generates the additional correction term characteristic of stochastic calculus.

Quadratic Variation and the Failure of First-Order Expansion

A fundamental difference between classical and stochastic calculus arises from the behavior of Brownian increments. In ordinary calculus, second-order infinitesimal terms vanish faster than first-order terms, so linear approximations are sufficient. For Brownian motion, however, the squared increment remains of order dt, leading to the key relation

$$(dW_t)^2 = dt \tag{3}$$

in the limit sense of quadratic variation. More precisely,

$$[W]_t = \lim_{n\to\infty} \sum_{j=0}^{n-1} \left(W_{t_{j+1}} - W_{t_j}\right)^2 = t \qquad \text{in probability as } \|\Pi\| \to 0 \tag{4}$$

almost surely.

Because $(\Delta W)^2 = \mathscr{O}(\Delta t)$ is of the same order as the drift, second-order terms in the Taylor expansion *survive* asymptotically and cannot be neglected. The additional correction term appearing in Itô's Lemma is precisely the consequence of this non-vanishing quadratic variation.

Taylor Expansion and Heuristic Rules

Let $f(t,x) \in C^{1,2}([0,T] \times \mathbb{R})$ be a sufficiently smooth function. The second-order Taylor expansion gives (the term $\frac{1}{2} f_{tt}(dt)^2$ vanishes since $(dt)^2 = 0$, not because $f_{tt} = 0$):

$$f(t+\Delta t, X+\Delta X) - f(t,X) \approx f_t \Delta t + f_x \Delta X + \tfrac{1}{2} f_{xx} (\Delta X)^2 \tag{5}$$

using the stochastic multiplication rules $(dt)^2 = 0$, $dt\, dW_t = 0$, $(dW_t)^2 = dt$, which eliminate all terms of order $(\Delta t)^{3/2}$ or higher. Substituting $\Delta X \approx \mu \Delta t + \sigma \Delta W$ and squaring:

$$(\Delta X)^2 = \mu^2 (\Delta t)^2 + 2\mu\sigma\, \Delta t\, \Delta W + \sigma^2 (\Delta W)^2 \;\longrightarrow\; \sigma^2 \Delta t \tag{6}$$

Therefore, applying the heuristic rules to the Itô process:

$$(dX_t)^2 = \sigma^2(t, X_t)\, dt \tag{7}$$

Substituting $\Delta t = dt$ and collecting terms, Itô's Lemma follows:

$$df(t,X_t) = f_t\, dt + f_x\, dX_t + \frac{1}{2} f_{xx} (dX_t)^2 \tag{8}$$

$$df(t,X_t) = f_t\, dt + \mu f_x\, dt + \sigma f_x\, dW_t + \frac{1}{2} f_{xx} \sigma^2\, dt \tag{9}$$

$$\boxed{df(t,X_t) = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x\, dW_t} \tag{10}$$

1.3 Geometric Brownian Motion as an Application of Itô's Lemma

Passing from the generic process X, we assume that asset price process follows, with instantaneous return form:

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t \qquad \Longrightarrow \qquad \frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t \tag{11}$$

To derive the explicit solution, apply Itô's Lemma to $f(t,S_t) = \ln(S_t)$. The partial derivatives are $f_t = 0$, $f_S = S_t^{-1}$, $f_{SS} = -S_t^{-2}$, which give directly:

$$d\ln(S_t) = \left(\mu - \tfrac{1}{2}\sigma^2\right) dt + \sigma\, dW_t \tag{12}$$

Integrating both sides from 0 to t and exponentiating, we obtain the Geometric Brownian Motion (**GBM**):

$$\ln\left(\frac{S_t}{S_0}\right) = \left(\mu - \tfrac{1}{2}\sigma^2\right) t + \sigma W_t \qquad S_t = S_0 \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right) t + \sigma W_t\right] \tag{13}$$

2. Derivation of the Black–Scholes PDE

2.1 Market Assumptions

The Black–Scholes framework rests on five idealizing assumptions:

- **Frictionless Markets:** No transaction costs, taxes, or bid-ask spreads.
- **Continuous Trading:** Portfolios are continuously rebalanced.
- **Constant Parameters:** The risk-free rate r and volatility σ are known and constant.
- **No Arbitrage:** Any self-financing riskless portfolio earns exactly r.
- **No Dividends (Baseline):** The underlying pays no dividends (we're gonna see it later).

2.2 Asset Dynamics

The underlying asset follows Geometric Brownian Motion GBM eq. (13) with constant drift μ and volatility σ. The risk-free bond evolves as $B_t = B_0 e^{rt}$, which implies the differential dynamic $dB_t = rB_t\,dt$.

2.3 The Replicating Portfolio and Self-Financing Condition

In order to derive the Black–Scholes partial differential equation, we first specify the market model and the notion of self-financing trading strategies.

Recall that for the market model $M = (B,S)$ a trading strategy is a pair $\varphi = (\varphi^1, \varphi^2)$ of progressively measurable stochastic processes defined on the underlying probability space $(\Omega, \mathscr{F}, \mathbb{P})$, which they represent respectively the number of share and bonds. A trading strategy $\varphi = (\varphi^1, \varphi^2)$ over the time interval $[0,T]$ **is said to be self-financing** if its wealth process $V(\varphi)$, of course without external resources, is defined by

$$V_t(\varphi) = \varphi_t^1 S_t + \varphi_t^2 B_t, \qquad \forall t \in [0,T], \tag{14}$$

satisfies

$$V_t(\varphi) = V_0(\varphi) + \int_0^t \varphi_u^1\, dS_u + \int_0^t \varphi_u^2\, dB_u, \qquad \forall t \in [0,T], \tag{15}$$

where the first integral is understood in the Itô sense and the second as a pathwise Riemann (or Lebesgue) integral.

In differential form, the self-financing condition reads $dV_t(\varphi) = \varphi_t^1\, dS_t + \varphi_t^2\, dB_t$.

Let $v(S_T, T)$ denote the $\mathscr{F}_T$-measurable payoff of a derivative security at maturity T. The payoff is path-dependent, meaning that it may depend not only on the terminal value S_T.

The central problem, is to determine a trading strategy (V_0, φ) such that an agent holding a short position in the derivative can perfectly hedge its payoff. In other words, we seek a self-financing strategy satisfying continuously in time.

$$V_T(\varphi) = v(S_T, T) \quad \text{almost surely.} \tag{16}$$

2.4 Delta Hedging and the No-Arbitrage Condition

Hedged portfolio. To construct a hedge, we build a portfolio φ consisting of one derivative and δ shares of the underlying, so that its value process at each time t is:

$$V_t(\varphi) = v(S_t,t) + \delta S_t. \tag{17}$$

Differential of the portfolio. Since v is a sufficiently smooth function on its domain, we apply Itô's Lemma to $dv(S_t,t)$:

$$dv(S_t,t) = \left(\frac{\partial v}{\partial t} + \mu S_t \frac{\partial v}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2}\right) dt + \sigma S_t \frac{\partial v}{\partial S_t} dW_t. \tag{18}$$

Adding the contribution from the δ shares via $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, the change in the value process $V_t(\varphi)$ is:

$$\begin{aligned} dV_t(\varphi) &= dv(S_t,t) + \delta\, dS_t \\ &= \left(\frac{\partial v}{\partial t} + \mu S_t \frac{\partial v}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2}\right) dt + \sigma S_t \frac{\partial v}{\partial S_t} dW_t \\ &\quad + \delta\,\mu S_t\,dt + \delta\,\sigma S_t\,dW_t. \end{aligned} \tag{19}$$

Collecting the dt and dW_t terms separately:

$$\begin{aligned} dV_t(\varphi) = &\underbrace{\left(\frac{\partial v}{\partial t} + \mu S_t \frac{\partial v}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2} + \delta\,\mu S_t\right)}_{\text{drift component}} dt \\ &+ \underbrace{\left(\sigma S_t \frac{\partial v}{\partial S_t} + \delta\,\sigma S_t\right)}_{\text{stochastic component}} dW_t. \end{aligned} \tag{20}$$

Elimination of market risk. To render $V_t(\varphi)$ instantaneously riskless, we require the coefficient of dW_t in (20) to vanish:

$$\sigma S_t \frac{\partial v}{\partial S_t} + \delta\,\sigma S_t = 0 \quad \Longrightarrow \quad \boxed{\delta = -\frac{\partial v}{\partial S_t}.} \tag{21}$$

This is the **Delta hedge**: at every instant the number of shares held equals minus the option's sensitivity to the underlying price, continuously adjusted as S_t evolves.

Substituting (21) into (20), the stochastic component vanishes and the μ-terms cancel exactly:

$$dV_t(\varphi) = \left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2}\right) dt. \tag{22}$$

No-arbitrage condition. Since $V_t(\varphi)$ is now locally riskless and self-financing, the no-arbitrage principle requires it to earn exactly the risk-free rate:

$$dV_t(\varphi) = rV_t(\varphi)\,dt. \tag{23}$$

Substituting $\delta = -\frac{\partial v}{\partial S_t}$ back into (17), the value process becomes:

$$V_t(\varphi) = v(S_t,t) - \frac{\partial v}{\partial S_t}S_t, \tag{24}$$

so the right-hand side of (23) expands as:

$$rV_t(\varphi)\,dt = r\left(v(S_t,t) - \frac{\partial v}{\partial S_t}S_t\right)dt. \tag{25}$$

Black–Scholes PDE. Equating (22) with (23):

$$\left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 v}{\partial S_t^2}\right)dt = r\left(v(S_t,t) - \frac{\partial v}{\partial S_t}S_t\right)dt, \tag{26}$$

and rearranging yields the **Black–Scholes PDE**:

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 v}{\partial S_t^2} + rS_t\frac{\partial v}{\partial S_t} - rv(S_t,t) = 0. \tag{27}$$

Two features of (27) deserve emphasis. First, the real-world drift μ has disappeared entirely: it cancelled at step (21) as a direct consequence of the Delta hedge. Second, the convective coefficient is r, not μ — under the no-arbitrage constraint the asset effectively drifts at the risk-free rate inside the PDE, independently of investor preferences.

2.5 Terminal, Boundary Conditions and Uniqueness

The Black-Scholes PDE is a universal equation; the specific type of derivative is uniquely determined by its terminal and boundary conditions. For a European call option C, with strike K and maturity T, these conditions are defined as follows.
There are two logic **Spatial boundary conditions.**

- **Lower boundary** ($S_t = 0$)**:** If the underlying asset price falls to zero, the call option becomes worthless:
$$C(0,t) = 0, \quad \forall t \in [0,T]. \tag{28}$$
- **Upper boundary** ($S_t \to \infty$)**:** As the stock price approaches infinity, the call is deep in-the-money and grows at the same rate as S_t, behaving like a forward contract:
$$\lim_{S_t\to\infty}\left[C(S_t,t) - \left(S_t - Ke^{-r(T-t)}\right)\right] = 0. \tag{29}$$

Terminal condition. At maturity ($t = T$), the value of the option must exactly match its terminal payoff:

$$C(S_T, T) = (S_T - K)^+. \tag{30}$$

2.6 The Feynman–Kac Theorem and the Black–Scholes Model

A variety of methods are available for solving the Black–Scholes partial differential equation (PDE). Among these, a particularly powerful approach is based on the Feynman–Kac representation, which provides a probabilistic interpretation of the solution and leads to a closed-form expression for option prices.

Let $V(t,S)$ be a sufficiently smooth function representing the price of a derivative security. In the Black-Scholes framework, V satisfies the parabolic PDE **Terminal condition:** At maturity T, the option value equals its payoff: $V(T,S) = H(S)$. The Feynman–Kac theorem states that the solution to the PDE for $t < T$ can be expressed as a discounted conditional expectation under the risk-neutral measure $\mathbb{Q}$:

$$V(t,S) = e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}[H(S_T) \mid \mathscr{F}_t]\,.$$

Here S_T denotes the stochastic process for the underlying asset price at maturity, and $\mathscr{F}_t$ represents the filtration$t$.

2.7 The Closed-Form Black-Scholes Formula

One way of solving the PDE is by transforming it into the heat equation yields the explicit analytical pricing formula for a European call option:

$$C(S_t,t) = S_t N(d_1) - Ke^{-r(T-t)}N(d_2), \tag{31}$$

where $N(x)$ denotes the cumulative distribution function of a standard normal. The components d_1 and d_2 are defined explicitly as

$$d_1 = \frac{\ln(S_t/K) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 = d_1 - \sigma\sqrt{T-t} = \frac{\ln(S_t/K) + \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \tag{32}$$

3. PART B: Continuous Dividend Yield

Motivation and Framework

The standard Black-Scholes derivation, as outlined in Part A, rests on the assumption that the underlying stock pays *no dividends* over the life of the option. This assumption can be relaxed following the extension proposed by Merton, who showed that the same PDE structure is preserved once the drift of the stock dynamics is adjusted to reflect the continuous leakage of value caused by dividend payments.

We consider a stock that pays dividends *continuously* at a constant proportional yield $q \geq 0$, so that the dividend paid over an infinitesimal interval dt is $qS_t\,dt$ per unit of stock held.

The stock price still follows a geometric Brownian motion, but the *total return* to the holder of the stock now has two components: capital gain dS_t and the dividend income $qS_t\,dt$.

Adjustment of the Hedging Portfolio

In the standard (no-dividend) Black-Scholes replication argument, a portfolio V is formed by holding one derivative $v(S_t,t)$ and $\varphi = -\partial v/\partial S$ units of the underlying stock (delta-hedging), so that the stochastic component is cancelled. When the stock pays a continuous dividend, *every share held in the portfolio generates an additional cash flow*. Because the hedger holds $-\Delta = -\partial v/\partial S$ shares, she *receives* (or pays, depending on the sign) the dividend stream

$$-qS_t\Delta dt \;=\; -qS_t\frac{\partial v}{\partial S}dt$$

over each interval dt. This extra term must be included in the self-financing condition. Accordingly, the instantaneous change in the portfolio value becomes

$$d\,V \;=\; dv(S_t,t) \;-\; \Delta dS_t \;-\; qS_t\Delta dt, \tag{33}$$

replacing the standard expression in which the last term is absent. The additional $-qS_t\Delta dt$ term reflects the fact that holding a short position in the stock obligates the hedger to pass through (or forgo) the dividend flow.

Derivation of the Modified PDE

Applying Itô's lemma to $v(S_t,t)$ exactly as in Part A yields

$$dv = \left(\frac{\partial v}{\partial t} + \mu S_t\frac{\partial v}{\partial S} + \tfrac{1}{2}\sigma^2 S_t^2\frac{\partial^2 v}{\partial S^2}\right)dt + \sigma S_t\frac{\partial v}{\partial S}dW_t.$$

Substituting into the modified portfolio equation and choosing $\Delta = \partial v/\partial S$ to eliminate the dW_t term, the portfolio becomes instantaneously risk-free and must therefore earn the risk-free rate r:

$$d\,V \;=\; r\,V\,dt.$$

Setting $V = v - \Delta S_t$ and equating both expressions for $d\,V$, one obtains—after cancellation of the μ-terms (exactly as in Part A) and collecting the additional $-qS_t(\partial v/\partial S)\,dt$ contribution—the **Black-Scholes PDE with continuous dividends**:

$$\boxed{\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 v}{\partial S^2} + (r-q)\,S_t\frac{\partial v}{\partial S} - r\,v \;=\; 0,} \tag{34}$$

which holds for $S_t > 0$ and $t \in [0,T)$, with the same terminal and boundary conditions as the standard European call (or put).

Economic Interpretation of the Drift Adjustment

The key structural difference between equation (34) and the standard Black-Scholes PDE lies in the coefficient of the first-order spatial derivative: the risk-free rate r is *replaced by* $r-q$.

This adjustment has a precise economic meaning. Under the risk-neutral measure Q, the total return on the stock—capital gain *plus* dividend—must equal the risk-free rate r. Since the stock already delivers a continuous dividend yield q, the required capital appreciation is only $r-q$. Formally, the risk-neutral dynamics of the stock become

$$dS_t \;=\; (r-q)\,S_t\,dt \;+\; \sigma S_t\,d\widetilde{W}_t,$$

where $\widetilde{W}_t$ is a Brownian motion under Q. The continuous dividend payment acts as a *drag* on the drift: the higher the yield q, the slower the risk-neutral capital appreciation of the stock, and consequently the lower the value of a call option (and the higher the value of a put option), everything else being equal.

Merton's Zero-Value Portfolio Approach

Merton derived the same PDE via a different but equivalent construction. He considered a self-financing portfolio ρ of N_v options, N_S shares and a cash position B_t, required to have *zero value* at every instant:

$$V_t = S_t N_S + v(S_t,t)\,N_v + B_t = 0.$$

When dividends are present, the cash account evolves as

$$dB_t \;=\; rB_t\,dt \;+\; \delta B_t \;+\; qS_t N_S\,dt, \tag{35}$$

where the last term captures the dividend income credited to the account for each share held. Following the same algebraic steps as in the no-dividend case— differentiating the zero-value constraint, invoking self-financing, eliminating risk via the choice $\varphi = \partial v/\partial S$, and applying Itô's lemma to v one arrives once again at equation (34). This confirms that the Merton extension is *robust to the choice of derivation approach*: both the replicating-strategy argument and the zero-value portfolio argument lead to the identical PDE, with the sole modification being the replacement of r by $r-q$ in the drift coefficient.

4. Conclusion

This project derived the stochastic foundations of modern derivatives pricing. Itô's Lemma provides the stochastic chain rule for functions of Brownian motion and produces the correction term arising from quadratic variation. Combining Itô's Lemma with a formally defined self-financing hedged portfolio and the strict no-arbitrage principle yields the Black–Scholes PDE, whose terms admit a clean economic interpretation in terms of Theta, Gamma, and the hedge ratio.

The Merton extension demonstrates that a continuous dividend yield q modifies only the drift coefficient, replacing the risk-free rate r with $(r-q)$ under the risk-neutral measure. By invoking the Feynman-Kac theorem, we can establish that these deterministic partial differential equations are mathematically equivalent to taking discounted expectations under the risk-neutral measure. Together, these concepts form the rigorous mathematical core of continuous-time option pricing.

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